RL for Abstractive Text Summarization[1]

Vera Monina

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Fundamental and Computational Linguistics

[1]Извините, пожалуйста,у нас на курсе принято, что можно писать на русском / английском; я подумала, что это надо было уточнить только, когда была уже на заключительном этапе. Надеюсь, что на английском тоже было. К тому же вся литература для этого реферата была на английском

Contents

Task Description

Classical Methods in ATS

Automatic text summarization (**ATS**) is a task of producing a concise, coherent representation of one or more source documents that preserves their most important content. It is one of the core problems in natural language processing (NLP). Over the past 30 years there were introduced statistical, graph-based methods and methods based on ML, DL and with an application of RL.

Two paradigms dominate the field. **Extractive summarization** selects and concatenates spans directly from the source, preserving literal phrasing. Despite interpretability, the extractive approach has fundamental limitation: mechanical extraction of sentences destroys anaphoric chains and does not provide the communicative coherence of the resulting summary.

Abstractive summarization generates novel text that may contain words and constructions absent from the original — much closer to how humans write summaries. This referat will focus on this type of the text summurization.

Linguistic criteria of summary quality

Evaluating the quality of automatically generated summaries is a research challenge in itself. Usually four basic dimensions are distinguished (Fabbri et al., 2021).

Coherence — a structural-discursive property: the summary must be organized as a coherent narrative. In terms of discourse linguistics, coherence is ensured by reference (coreference), discourse connectors, and adherence to the principles of information structure — the distinction between given and new. Violations of coherence manifest as broken anaphoric chains, unmotivated topic shifts, and disorders in Rhetorical Structure Theory.

Consistency — the correspondence of the summary's content to the actual information of the source document. Linguistically, this is operationalised through textual entailment: each statement in the summary must follow from the source text as a premise. The main problems here are special type of violations — hallucinations — that now are widespread across all new LLM's (and agent systems).

Fluency — the quality of individual sentences.

Relevance — the correspondence of the summary's content to the most significant information of the source, related to the notion of informativeness.

A Brief History: Before Reinforcement Learning

The earliest ATS systems, dating to Luhn (1958) and Edmundson (1969), scored sentences by term frequency and positional heuristics. Baxendale (1958) showed empirically that the first and the last sentences carry the highest informational load. Graph-based methods — TextRank (Mihalcea & Tarau, 2004) ranked sentences by centrality in similarity graphs, without any supervision. However it was long ago, but such graph-methods are still relevant for some linguistic-based goals, like comparison of standard and non-standard essays.

The neural era began with Rush et al. (2015), who adapted an attention-based encoder–decoder (seq2seq) to title generation. Nallapati et al. (2016) cast extractive summarization as sequence labelling with RNNs. Liu & Lapata (2019) showed that BERT-based encoders substantially outperform earlier extractors. Lewis et al. (2020) proposed **BART** — a pre-trained encoder–decoder — which became the baseline of text summurization for several years.

These systems shared a common training objective: **supervised maximum-likelihood estimation (MLE)** against reference summaries (produced by professional annotators), evaluated with **ROUGE** (Lin, 2010). This created a fundamental tension: a model trained to predict gold tokens token-by-token is evaluated at inference time on full sequence quality and judged by humans.

Why MLE Is Not Enough

As it was found out, pre-RL abstractive systems suffered from three structural problems:

Exposure bias: during training, the model learns from correct previous words. But when generating text later, it has to rely on its own possibly wrong predictions — a situation it never experienced during training.

Metric–objective mismatch: cross-entropy over tokens does not correlate well with ROUGE, and neither correlates well with human preference.

Factual hallucinations: models frequently generate plausible-sounding but factually incorrect statements (as we mentioned before).

These difficulties motivated the turn to reinforcement learning.

Why RL Is Necessary

The core motivation is that the true objective — a human's judgement of summary quality — is **non-differentiable and non-decomposable into token-level targets**. RL provides a framework for optimising any scalar feedback signal, including human preference scores or automatic evaluators that assess full-sequence properties.

Architecture and Implementation of RL for Summarization

Task Formalization (MDP)

Following the RL framework, the Abstractive Summarization (AST) task is modeled as a Markov Decision Process (MDP) as follows[2]:

Agent	The language model (policy π_φ) that generates the summary text token by token.
Environment	The source document space $\mathcal{X}$ combined with the frozen Reward Model (RM). It is partially observable: the agent sees the prefix but receives feedback only at the end.
Action	Selection of the next token y_t from the vocabulary $\mathcal{V}$ ($\lvert\mathcal{V}\rvert \approx +50,000$).
Reward	Sparse and terminal. It is calculated only after the `EOS` token is generated, using the RM based on human preference alignment (usually).

Listing 1: MDP components for RL-based summarization.

1. Algorithmic Chain (Pipeline)

The lifecycle of the system is divided into three critical stages: training, inference, and production deployment.

[2]Such models are also known as RLHF (hf - human feedback)

1.1. Training (Three-Stage Pipeline)

a. **SFT (Supervised Fine-Tuning):** The model is trained on (document (one and more), reference) pairs using Cross-Entropy to create a strong baseline.

b. **RM (Reward Modeling):** A classifier is trained on human-ranked outputs ($y_w > y_l$) using the Bradley-Terry loss:

$$\mathcal{L}_{\text{RM}} = -\log \sigma(r_\theta(x, y_w) - r_\theta(x, y_l))$$

c. **RL (PPO Phase):** The policy π_φ is optimized to maximize r_θ while applying a KL-divergence penalty to prevent the model from drifting too far from the SFT initialization.

1.2. Inference

Decoding: Beam Search or Top-p (Nucleus) Sampling.

Optimization

2.3. Production

Efficiency: Use model quantization (INT8/FP8) and distillation to lower inference costs.
Monitoring the resuts

Two ACL Architectures in Focus

(ACL – is a leading conference for computational linguists)
We examine two approaches published at ACL from 2023 to 2024, that apply RL to ATS — both use the PPO-based RL pipeline above, but differ in how the reward is constructed.

Algorithm 1 (MDO) — Ryu et al. (2024) propose multi-objective RL using four UniEval dimensions as simultaneous rewards.

Algorithm 2 (RLEF) — Roit et al. (2023) use a single NLI-based entailment reward targeting factual consistency.

Both are described in this Section.

Algorithm 1: Multi-Dimensional Optimization (MDO) — Ryu et al., ACL 2024

Reward Design

Instead of optimising a single scalar reward, MDO defines a **four-dimensional reward vector** (criteria included the linguistic aspects that we observed previously):

$$R(y) = \left[r_{\text{coh}(y)},; r_{\text{con}(x,y)},; r_{\text{flu}(y)},; r_{\text{rel}(x,y)}\right] \in \mathbb{R}^4$$

Each component is computed by **UniEval**, a QA-based multi-dimensional evaluator that closely aligns with human preferences. Unlike ROUGE, UniEval does not require reference summaries.
The final reward sent to PPO is obtained via one of two MDO strategies[3]:

MDO_min (robust optimisation): reward the current worst-performing dimension:

$$R_{\text{scalar}(y)} = \min_i r_{i(y)}$$

[3]There may be errors in the indexes in formulas because not everything has been compiled correctly (sorry)

$$\nabla_\theta J \approx R_{\text{scalar}(y)} \cdot \sum_t \nabla_\theta \log \pi_\theta(a_t \mid s_t)$$

MDO_pro (gradient projection / PCGrad): for each objective i, compute gradient $g_i = \nabla_\theta \mathbb{E}\left[r_{i(y)}\right]$. When two gradients conflict ($g_i \cdot g_j < 0$), project:

$$g^* = \arg\min_g \left\| g - \sum_i \alpha_i g_i \right\|^2 \quad \text{s.t.} \quad g \cdot g_i \geq 0 \quad \forall i$$

$$\theta \leftarrow \theta + \eta g^*$$

MDO_min is simpler and more interpretable. MDO_pro is more stable under conflicting objectives at the cost of extra computation per step.

Architecture Diagram

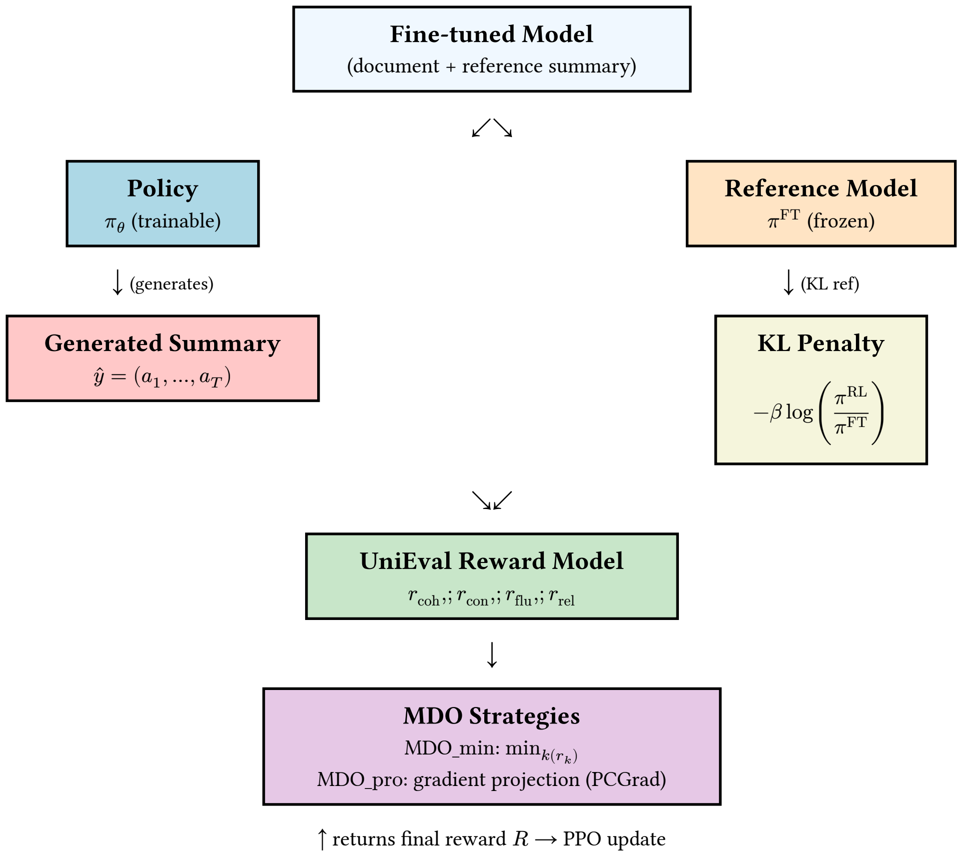

Figure 1: Multi-dimensional Optimization pipeline (Ryu et al., ACL 2024).

Training Pipeline

1. **Initialise**: Load SFT model as both policy π_θ (trainable) and reference π^{FT} (frozen).
2. **Sample**: Draw document $x \sim \mathcal{D}$; rollout full summary $\hat{y} = (a_1, ..., a_T) \sim \pi_{\theta(\cdot|x)}$.
3. **Score**: Compute 4 UniEval dimensions: $[r_{\text{coh}}, r_{\text{con}}, r_{\text{flu}}, r_{\text{rel}}]$.
4. **Aggregate**: Apply MDO_min or MDO_pro to obtain scalar R.
5. **KL**: Add penalty $-\beta \log\left(\frac{\pi^{\text{RL}(\hat{y}|x)}}{\pi^{\text{FT}(\hat{y}|x)}}\right)$.
6. **Update**: Compute PPO clipped loss; backpropagate; update θ.
7. **Repeat** until convergence or KL budget exhausted.

Results presented in the article

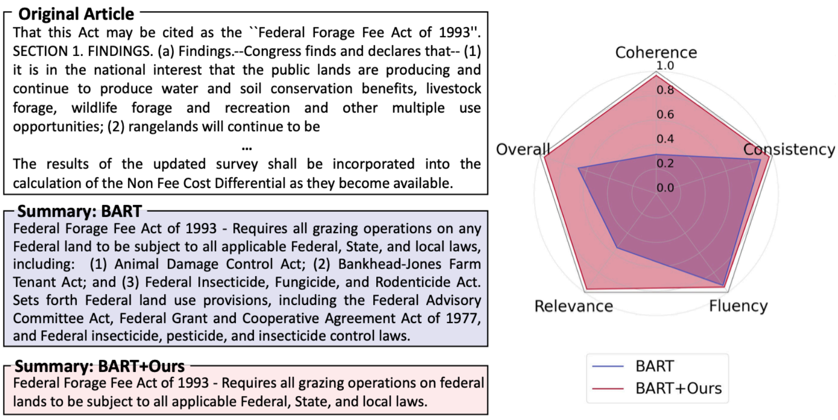

Figure 1: While the baseline model produces an imbalanced summary (■), we aim to generate overall high-quality summaries (■). The radar chart illustrates `UniEval` scores for four dimensions.

BART – is a state-of-the-art model, that we observed earlier (encoder-decoder model)

Algorithm 2: RLEF — RL with Entailment Feedback — Roit et al., 2023

Motivation

RLEF targets a single but critical dimension: **factual consistency** (consistency/faithfulness). Rather than human preference pairs, the reward signal is derived from an **NLI model**: the probability that the source document entails the generated summary.

Reward Design

$$r_{\mathrm{NLI}(x,y)} = p_{\mathrm{NLI}(\text{entailment} \mid x,y)}$$

The full reward mirrors standard RL:

$$R(x,y) = r_{\mathrm{NLI}(x,y)} - \beta \cdot \mathrm{KL}\left(\pi_{\varphi} \parallel \pi_{\mathrm{SFT}}\right)$$

No human annotation is needed at RL of this type — the NLI model acts as a frozen automatic reward model. So this is one of the first algorithms that goes from human annotations.

Architecture Diagram

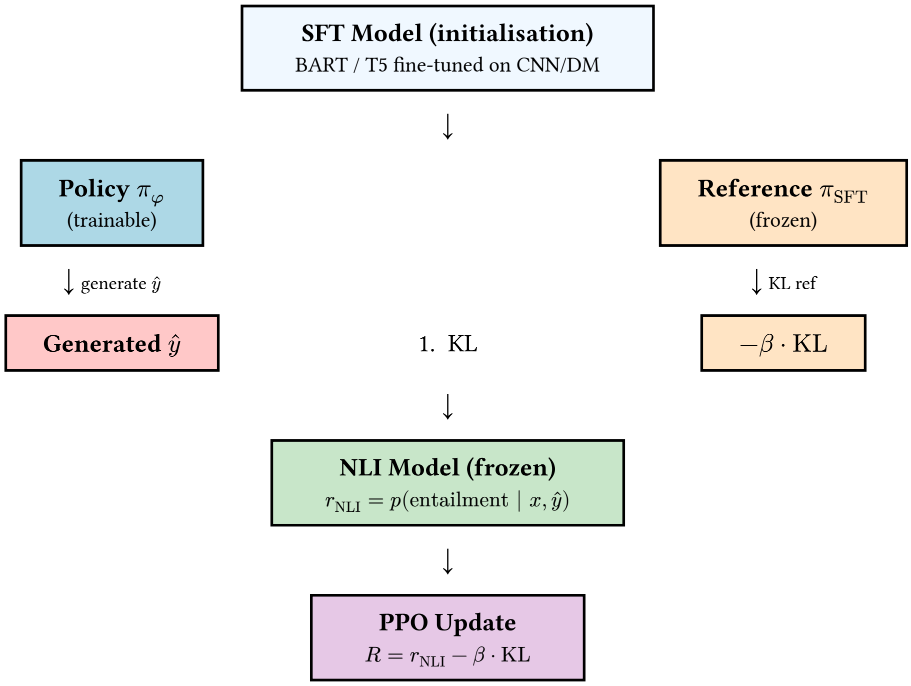

Figure 3: RLEF pipeline (Roit et al., ACL 2023).

Training Pipeline

1. **Initialise**: SFT model $\longrightarrow$ policy π_φ (trainable) and reference π_{SFT} (frozen).
2. **Rollout**: Generate full summary $\hat{y} \sim \pi_{\varphi(\cdot|x)}$.
3. **NLI reward**: $r_{\text{NLI}} = p_{\text{NLI(entailment} \mid x, \hat{y})}$ from frozen NLI model.
4. **Penalty**: Subtract $\beta \cdot \text{KL}\left(\pi_\varphi \parallel \pi_{\text{SFT}}\right)$.
5. **PPO update**: Standard clipped surrogate objective.
6. **Extractiveness guard**: An additional ROUGE-overlap penalty term discourages verbatim copying (extractive collapse artefact).

Examples of generated summaries from the article:

	Summary before the NLI flip	Summary after the NLI flip	NLI	Description
1	Two astronauts who spent a year living on the International Space Station have landed in Florida.	Two astronauts who spent a year living on the International Space Station have returned to Earth.	✓	Abstractive Rephrasing
2	Afghan forces have repelled an advance by Taliban fighters on the northern city of Kunduz, officials say.	Afghan forces have been battling Taliban insurgents in the northern city of Kunduz.	✓	Abstractive Rephrasing
3	A senior Nigerian military official has said militant Islamist group Boko Haram is no longer a threat, after a mosque attack that left at least 82 people dead.	A senior Nigerian official has denied that Islamist militant group Boko Haram was behind a mosque attack in the north in which more than 100 people were killed.	✓	Claim Change
4	Two people have been arrested on suspicion of manslaughter after a three-year-old boy died at a water park.	Two people have been arrested after a four-year-old boy died at a water park.	✓	Argument Omission
5	Bolton Wanderers manager Lee Trotter has apologised after he and striker Aaron Lennon swore at fans on live television.	Bolton Wanderers manager Lee Trotter has apologised after he and team-mate Gary Caldwell swore at fans on live television.	✗	Argument Change

Table 4: Examples of summaries for the same document on consecutive checkpoints during RL training, before and after the NLI classification of the summary has flipped. The summaries maintain a stable main structure which enables manual inspection. The main changes are highlighted in gray, the NLI column depicts the entailment decision for the latter summary, and the description specifies the type of the semantic change.

Challenges and Potential Issues (Corner Cases)

Implementing Reinforcement Learning (RL) for Abstractive Text Summarization still introduces several structural and linguistic challenges:

1. **Reward Hacking:** The model finds "cheats" to get a high score from the Reward Model (RM) without actually writing a good summary.

2. **Language Collapse:** If we push the RL training too hard, the model might start "breaking" and producing repetitive or nonsensical text.
3. **Extractive Collapse:** Sometimes, the model gets lazy and just copies big chunks of text from the source to make sure it stays "factually consistent." This ruins the whole point of an **abstractive** summary.
4. **Factual Hallucinations:** Even with RL, models still struggle with "hallucinations".
5. **Subjectivity and Bias:** Since in the majority of works uses the reward comes from humans, the model learns from biased data. Agreement between human annotators is only around 72% (from my experience usually even lower), meaning the RM might pick up the specific cultural or stylistic "vibes" of a small group of people instead of objective facts.
6. **Reward Overoptimization:** This is a tricky one. The longer you train, the higher the Reward Model (RM) score goes, but actual quality starts to drop after a certain peak. It's very hard to know exactly when to stop training automatically.
7. **High Computational Cost:** RL is a massive three-stage process (SFT, RM, and PPO). PPO also requires an extra "critic" model to work. This makes the whole thing at least 2–3 times more expensive and slower than standard fine-tuning.

Conclusion and Final Thoughts

The paradigm shift toward Reinforcement Learning from Human Feedback (RLHF) has fundamentally redefined abstractive summarization. We have moved from simple token-level prediction (MLE) to holistic sequence optimization based on human-aligned values.

Current Technological Standing

State-of-the-Art: PPO-based pipelines, such as MDO and RLEF, currently represent the "gold standard" for performance in computational linguistics.
Methodological Necessity: RL is required because high-level linguistic properties—such as coherence, consistency, and relevance—cannot be reduced to differentiable token-level targets.
Evolution of Metrics: There is a definitive trend toward reference-free evaluators (e.g., UniEval) that align more closely with human judgment than traditional n-gram overlap metrics like ROUGE or token overlap metric like Cross-Entropy.

Limitations and Linguistic Gaps

While current RL approaches improve communicative success, they remain limited from a deeper linguistic perspective:
The "Linguistic Ceiling": Current training methods focus only on big-picture qualities like how fluent or logical the text sounds, not on finer details (like factual accuracy or style nuances), but fails to address fundamental differences in lexical density, syntactic variation (e.g., part-of-speech distribution), and semantic depth found in human-authored text.
Reward Overoptimization: Extensive training often leads to a "peak" in RM scores followed by a decline in actual linguistic quality—a phenomenon that currently lacks an automatic detection mechanism.
Computational and Subjective Barriers: The high cost of PPO (requiring multiple model copies) and the inherent subjectivity of human reward signals (72% inter-annotator agreement) introduce noise into the learning process.
Data Contamination & Evaluation Validity: Recent work in computational linguistics highlights a critical but often overlooked issue: for a valid assessment of a model's true linguistic understanding (and for the correct interpretation), we must strictly ensure that no part of the evaluation data appears in the training set. However, this is extremely difficult to guarantee for large proprietary models (e.g., Yandex, GPT, Claude, Gemini, DeepSeek) because their training data is rarely fully closed for general users.
This problem is especially serious for ATS benchmarks. CNN/DailyMail and Reddit — are one of the main datasets used in both MDO and RLEF — are among the oldest and most widely distributed NLP corpora. They have been publicly available since 2015–2017 and were almost certainly included in the pretraining data of BART, GPT-3. This means that when we evaluate a model on CNN/DailyMail, we may be measuring memorisation rather than actual summarization ability.
The Register and Style Gap: All existing RL approaches optimise toward a single averaged style — whatever the annotators preferred. But summarization in practice is stylistically diverse: an academic abstract, a news lead, a legal

annotation follow very different lexical, syntactic, and discourse norms. Neither MDO nor RLEF model register as a quality dimension. A reward model trained on Reddit will systematically score formal academic summaries lower.
Discourse Structure Is Not Rewarded: UniEval measures coherence as a global property — whether the text reads well overall. But in discourse linguistics, coherence is built from specific mechanisms: coreference chains, discourse connectors, rhetorical relations (RST) as I said before. None of the four UniEval dimensions check whether anaphoric chains from the source are preserved, whether pronouns are resolved correctly.

Future Directions

While methods like DPO (Direct Preference Optimization) offer more efficient training, the next frontier in AST must involve a more granular linguistic approach. Here are some of them:
Syntactic Probes as Reward Signal: A concrete next step would be integrating syntactic probing classifiers directly into the reward function. For example, a dependency parser could verify that the argument structure of predicates is preserved. A constituency parser could flag broken NP/VP[4] structures that are typical of hallucinations. This is technically easy to integrate but has not been done yet (even there has been already invented rule-based and neural parsers).
Cross-Lingual Summarization: Both observed algorithms were trained and evaluated on English. For morphologically rich languages (Russian, Arabic, Finnish) or languages with flexible word order, the exposure bias problem is significantly worse: the space of valid continuations is larger, and early token errors more severely damage syntactic structure. Extending MDO and RLEF to such languages would require language-specific reward components — for example, morphological agreement[5] as a separate quality dimension.
Minimal Pairs as a Reward Diagnostic: Instead of evaluating summaries randomly, one could construct minimal pairs — a summary with one targeted factual substitution (similar to RuBLiMP (Taktasheva E. et al., 2018)) — and check whether the reward model distinguishes them. If it does not, the reward model is insensitive to factual errors and responds only to surface style. This would be a new linguistically controlled way to diagnose reward model quality, directly connecting summarization evaluation to the probing methodology used in interpretability research.

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[4]NP - noun phrase: вершина “словосочетания” существительное; VP - verb phrase: вершина “словосочетания” глагол
[5]соглосование: вкуснЫЙ пирог, красивАЯ девочка